14 SQL patterns, one mental map

Identify the pattern first. Then write the query.

01 Window ranking

Rank rows within groups

TOOL: ROW_NUMBER(), RANK(), DENSE_RANK()

WATCH: Define how ties should behave.

02 Top N by group

Return leaders in each segment

TOOL: PARTITION BY group_key + rank filter

WATCH: Choose ROW_NUMBER or DENSE_RANK intentionally.

03 Running totals

Accumulate values over a timeline

TOOL: SUM(amount) OVER (ORDER BY dt)

WATCH: Specify the window frame for tied dates.

04 Previous / next

Compare adjacent observations

TOOL: LAG(metric) / LEAD(metric)

WATCH: Partition separately for each entity.

05 Self joins

Relate rows within one table

TOOL: JOIN employees e ON e.manager_id = m.id

WATCH: Use LEFT JOIN when roots must stay.

Think in reusable patterns

Recognize the cue; choose the right SQL tool.

06 Anti-joins

Find missing relationships

TOOL: NOT EXISTS (SELECT 1 FROM ...)

WATCH: NOT IN + NULL can surprise you.

07 Grouped aggregates

Summarize one row per group

TOOL: GROUP BY + COUNT/SUM/AVG + HAVING

WATCH: Filter rows before GROUP BY with WHERE.

08 Conditional pivots

Turn categories into columns

TOOL: SUM(CASE WHEN ... THEN ... END)

WATCH: Include a sensible ELSE and NULL policy.

09 Deduplication

Keep one preferred row per key

TOOL: ROW_NUMBER() OVER (PARTITION BY key...)

WATCH: Make the tie-break order deterministic.

10 CTE pipelines

Name intermediate query steps

TOOL: WITH step_a AS (...), step_b AS (...)

WATCH: CTEs improve clarity, not always speed.

Interview-ready SQL reasoning

Cover edge cases, not just the happy path.

11 Correlated subqueries

Test a condition per candidate

TOOL: EXISTS (SELECT 1 ... WHERE outer.id=...)

WATCH: Inspect execution plans before claiming slower.

12 Gaps and islands

Group consecutive dates / IDs

TOOL: value - ROW_NUMBER() as group key

WATCH: Watch for duplicate values and date gaps.

13 Date cohorts

Measure activity after a start date

TOOL: DATE_TRUNC + first_event + GROUP BY

WATCH: Handle time zones and partial periods.

14 Recursive queries

Walk hierarchical relationships

TOOL: WITH RECURSIVE nodes AS (...)

WATCH: Bound depth and guard against cycles.

MINI CHECKLIST

Clarify NULLs and ties | Pick a grain | Verify joins and filters
Test small examples | Explain assumptions | Check the execution plan